

2023

MATHEMATICS — HONOURS

Paper : CC-10

(Mechanics)

Full Marks : 65

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*

1. Answer the following multiple choice questions with only one correct option. Choose the correct option with proper justification : (1+1)×10

- (a) If X, Y be the algebraic sum of the resolved parts of a system of coplanar forces along the rectangular axes OX and OY respectively and G be the algebraic sum of the resolved parts of the moments the forces about O , then the locus of the point about which the algebraic sum of the moments of the forces is equal to a constant G' , is given by :

(i) $Yx - Xy = G' - G$

(ii) $Yx - Xy = G - G'$

(iii) $Yy - Xx = G' - G$

(iv) $Yy - Xx = G - G'$

- (b) Suppose a force F is acting along the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$, then the moment of F about z -axis will be

(i) $F(x_1m - y_1l)$

(ii) $F(y_1n - z_1m)$

(iii) $F(z_1l - x_1n)$

(iv) $F(z_1l - y_1n)$

- (c) A lamina in the form of a cycloid rests on the top of another cycloidal lamina. The radii of the generating circles of the upper and lower cycloid are ' a ' and ' b ' respectively. If ' h ' is the height of the c.g. of the upper cycloid above its vertex, then the equilibrium will be stable if :

(i) $h > \frac{2ab}{a+b}$

(ii) $h < \frac{2ab}{a+b}$

(iii) $h > \frac{4ab}{a+b}$

(iv) $h < \frac{4ab}{a+b}$

- (d) If the resultant R of two forces $2P$ and P is the geometric mean between the greatest and the least values of R , then the angle between the forces P and $2P$ is

(i) $\frac{\pi}{2}$

(ii) $\frac{\pi}{6}$

(iii) $\frac{2\pi}{3}$

(iv) $\frac{5\pi}{6}$

Please Turn Over

- (c) The potential energy U of a conservative system varies with distance x and $U = 3x^2 - 2x$. Then the force acting on the system
- (i) is proportional to x (ii) is proportional to x^2
(iii) decreases constantly (iv) decreases linearly with x .
- (f) A man starts walk at 4 miles per hour and his speed at any instant is inversely proportional to $(x + 10)$, where x is the number of miles he has walked. Then the time he takes to walk 20 miles is
- (i) 10 hours (ii) 15 hours
(iii) 20 hours (iv) None of these.
- (g) If the radial velocity is proportional to the transverse velocity of a particle moving in a plane curve, then the plane curve is
- (i) a parabola
(ii) a hyperbola
(iii) a cardioid
(iv) an equiangular spiral.
- (h) A particle describes the path $p^2 = ar$ under a force F to the pole. Then the law of force will be
- (i) $F \propto \frac{1}{r}$ (ii) $F \propto \frac{1}{\sqrt{r}}$
(iii) $F \propto \frac{1}{r^2}$ (iv) $F \propto r^2$.
- (i) If the ratio of the elliptical orbits of two planets is $\frac{9}{16}$, then the ratio of their periodic time is
- (i) $\frac{3}{4}$ (ii) $\frac{4}{9}$
(iii) $\frac{8}{27}$ (iv) $\frac{27}{64}$.
- (j) A uniform hollow cylinder (circular) is rolling down a rough plane inclined at an angle α to the horizon. Then the kinetic energy of the cylinder at any time t is given by
- (i) Mv^2 (ii) $\frac{1}{2}Mv^2$
(iii) $\frac{1}{3}Mv^2$ (iv) $\frac{1}{4}Mv^2$,
- where M is the mass of the cylinder and v is the velocity of its centre of mass at any time t .

(3)

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Unit - 1

(Marks : 05)

2. Answer *any one* question :

- (a) Two uniform similar rods of same materials PQ and QT of length $2l$ and $2l'$ respectively are rigidly united at Q and suspended freely from P . If they rest inclined at an angles α and β respectively to the vertical, prove that $(l^2 + 2l'l') \sin \alpha = l'^2 \sin \beta$. 5
- (b) OA, OB, OC are three co-terminous edges of a cube and AA', BB', CC' and OO' are diagonals. Along BC', CA', AB' and OO' act forces equal to X, Y, Z and R respectively. Show that they are equivalent to a single resultant if $(YZ + ZX + XY)\sqrt{3} + R(X + Y + Z) = 0$. 5

Unit - 2

(Marks : 05)

3. Answer *any one* question :

- (a) Four equal rods each of weight W form a rhombus $ABCD$ with smooth hinges at the joints. The frame is suspended by the end A and a weight W' is attached at C . A stiffening rod of negligible weight joins the middle points of AB, AD keeping these inclined at an angle α to AC . Show that the thrust in the stiffening rod is $(4W + 2W') \tan \alpha$. 5
- (b) A beam of length ' l ' rests with its ends on two smooth planes which intersect in a horizontal line. If the inclination of the planes to the horizon are α and β and the c.g. of the beam divides it in the ratio $a : b$; find the position of equilibrium of the beam and show that it is unstable. 5

Unit - 3

(Marks : 10)

4. Answer *any two* questions :

- (a) Deduce the radial and cross-radial components of velocity and acceleration of a particle moving in a plane. 5
- (b) A shot of mass ' m ' penetrates a thickness ' s ' into a fixed beam of mass M . Prove that if M is free to move, the thickness penetrated is $\frac{Ms}{M + m}$. 5
- (c) If a particle moves from one position to another position on a smooth plane curve under the action of conservative forces in the plane, then show that the change in kinetic energy is equal to the work done by the forces. 5
- (d) An engine is pulling a train starting from rest, works at a constant rate H units per second. If M is the mass of the train and the frictional resistance is $K \times (\text{velocity})^2$, find the distance when the train will obtain velocity u . Also find the maximum speed of the train. 5

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Unit - 4

(Marks : 15)

5. Answer **any one** question :

- (a) A particle is thrown upwards with speed V . If the air resistance is assumed to vary as the square of the speed and the resistance is equal to gravity (supposed constant) when the speed is U , show

that the particle will rise for a time $\frac{U}{g} \tan^{-1} \left(\frac{V}{U} \right)$. Also prove that the velocity with which it returns

to the point of projection is $\frac{UV}{(U^2 + V^2)^{\frac{1}{2}}}$. 4+3

- (b) A point is moving in a straight line with an acceleration μx towards a fixed centre in the straight line and with an additional acceleration $L \cos pt$, where μ , L and p are positive constants and

x is the distance from the fixed point. If $p = \sqrt{\mu}$ and $x = a$, $\frac{dx}{dt} = 0$, when $t = 0$, find x and $\frac{dx}{dt}$ at any time t . 7

6. Answer **any one** question :

- (a) A particle moves under a central acceleration $\frac{\mu}{r^3}$. It is projected from an apse at a distance ' a ' from the centre of force with a velocity equal to $\sqrt{2}$ times the velocity in a circle at the same distance. Prove that the path is $r \cos \left(\frac{\theta}{\sqrt{2}} \right) = a$. 8

- (b) A particle projected along the inner surface of a rough sphere is acted on by no forces. Show that it will return to the point of projection at the end of time $\frac{a}{\mu v} (e^{2\pi\mu} - 1)$, where ' a ' is the radius of the sphere, ' v ' is the velocity of projection and ' μ ' is the coefficient of friction. 8

Unit - 5

(Marks : 10)

7. Answer **any one** question :

- (a) (i) If two smooth spheres of masses m_1 and m_2 moving with respective velocities u_1 and u_2 in the same direction impinge directly, find (i) their velocities after impact and (ii) loss of K.E.

- (*) A ball is thrown from a point on a smooth, horizontal ground with a velocity V at an angle α to the horizon. Show that the total time for which the ball rebounds on the ground is

$$\frac{2V \sin \alpha}{g(1-e)} \text{ and that its distance from the starting point when it ceases to rebound is } \frac{V^2 \sin 2\alpha}{g(1-e)},$$

5+5

e being the coefficient of elasticity between the ball and the ground.

- (b) (i) Define linear momentum for a multiparticle system consisting of particles with masses $m_1, m_2, \dots, m_N$ and velocities $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_N$. Interpret it in terms of motion of centre of mass of the system.
- (ii) Write down linear momentum principle for such multiparticle system and interpret it in terms of an isolated system.
- (iii) The position of a particle P of mass ' m ' at time t is given by $x = at, y = 2at, z = 3at$. Find angular momentum of P about a point $A(a, a, a)$ and about the axis of z at any time t .

(1+1)+(2+2)+(2+2)